

Syllabus Breakdown

(15 sessions approx)

1. Fundamentals of device drivers
2. Linux file hierarchy
3. Linux Kernel Architecture (Brief)
4. Writing first Linux device-driver
5. Building (native) & Testing
6. Introduction of Raspberry Pi3
Model B
7. Installation of Raspbian OS &
Connecting Raspberry Pi to PC
8. Kernel module dependencies
9. Building own customised Linux for
RPi3 using Buildroot tool
10. Buildroot customization &
compilation
11. Kernel customization & porting
12. Character device driver
13. Kernel space and user space
communication
14. IOCTL
15. IO driver
16. Project

Key Takeaways from the Course

1. Kernel space programming skills
2. Setting up cross compilation environment in your Linux machine
3. Native & cross compilation of Linux device drivers
4. Static and dynamic compilation of Linux device driver
5. Building your own customized Linux for your target SBC
6. Hands-on IO driver

Prerequisites for learning Linux device driver:

- C + DS
- Linux user space programming
- Microcontroller/SoC programming
- Shell scripting

Hardware requirement for practicing Linux device driver

- Ubuntu 18.04
- Any SBC (eg: Raspberry Pi)
- RJ45 Ethernet LAN network cable
- USB to Serial converter
- Some other necessary hardware based upon requirement